Mid Semester Examination 2026

Institute	IITE	Exam Month / Year	March 2026
Program	B.Tech	Branch	CE/CSE/IT
Semester	4	Regular / Repeater	Regular
Course Code	CE0417	Course Name	Data Structure & Algorithm
Date	28/03/2026	Time	1:00 PM to 2:30 PM
Day	Saturday	Maximum Marks	40

Instructions :

Attempt all questions.
Make suitable assumptions wherever necessary.
Figures to the right indicate full marks.

Q 1 a) What is a quick sort? Write down a quick sort algorithm. **04**

b) What is the essential condition for binary search to apply on any data? Write an algorithm of Binary search for searching a given data. **04**

OR

b) Define Data Structure and differentiate between linear and nonlinear data structures. **04**

Q 2 a) Write down the algorithms for PUSH and POP operations. **04**

b) Convert Infix Expression (A + B) * C – D ^ E ^ (F * G) into prefix format showing stack status after every step in tabular form. **06**

OR

b) Write an algorithm to insert a node at the beginning of the singly linked list and after the specific position in singly linked list. **06**

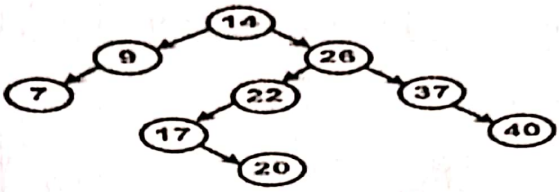
c) What is a circular queue? Also write an algorithm for insertion and deleting an element from a circular queue. **06**

OR

c) Write an algorithm to delete a node at the beginning and at the end of the circular linked list. **06**

Scanned with OKEN Scanner

Q 3 a) Perform the following operations in order on the given binary search tree, and draw the tree after performing each operation: first insert 10, then insert 24, then delete 37, and finally delete 22. 04



b) Given Inorder and Preorder traversal, find Postorder traversal: 06

Inorder: B F G H P R S T W Y Z

Preorder: P F B H G S R Y T W Z

OR

b) Construct an AVL tree having the following elements 06

20,11,5,32,40,2,4,27,23,28,50

c) Find the total weight of the minimum spanning tree with step by step explanation using Krushkal Algorithm.(refer figure (a)) 06

OR

c) Find the total weight of the minimum spanning tree with step by step explanation using Prim's Algorithm.(refer figure (a)) 06

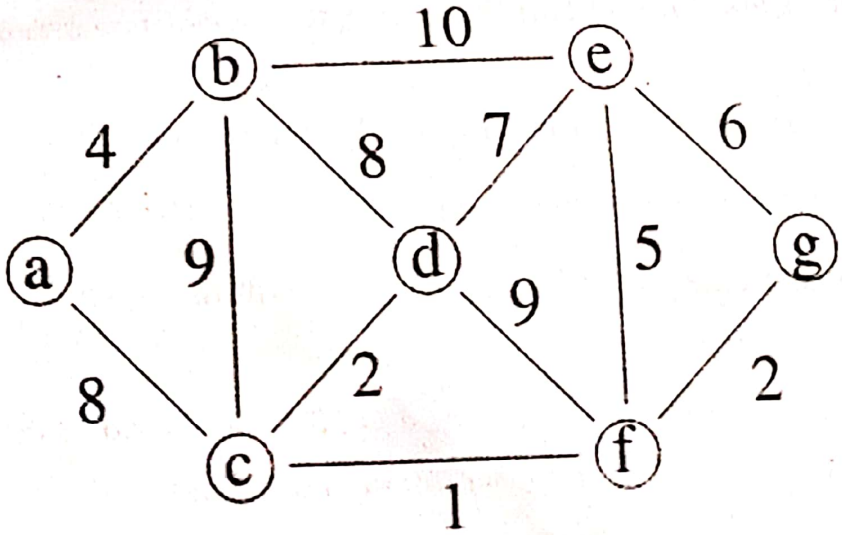


Figure (a)

Scanned with OKEN Scanner